

Accelerating Design Exploration and Optimization with Surrogate Physics Models

B.Tech Project Presentation

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Overview

- 1 Introduction
- 2 Methodology
- 3 Results
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Motivation & Problem Statement (Recap)

The Challenge:

- Design optimization requires high-fidelity simulations (CFD/FEA).
- Single simulations take hours/days; iterative design is computationally expensive.
- Additive Manufacturing enables complex geometries (e.g., Gyroids), vastly expanding the design space.

The Solution:

- **Surrogate Modeling:** Use Machine Learning (Gaussian Process Regression) to approximate physics.
- **Inverse Design:** Maximize performance subject to engineering constraints.

The Gyroid Heat Exchanger (Recap)

Why Gyroids?

- Triply Periodic Minimal Surface (TPMS).
- High surface-to-volume ratio.
- Continuous channels (no dead-ends).
- Use case : Aerospace, Automotive engines



Figure: Gyroid Heat Exchanger

Optimization Objective

Maximize Surface Area (A_s) subject to:

- **Mass:** $M < 125 \text{ g}$
- **Pressure Drop:** $\Delta p < 8000 \text{ Pa}$
- **Avg Velocity:** $v_{avg} > 520 \text{ mm/s}$

Simulation: Lattice Boltzmann Method (LBM)

We utilize LBM for creating the training dataset.

- **Solver:** Solves discrete Boltzmann equation on a voxel grid.
- **Boundary Condition:** Bounce-back scheme (handles complex gyroid boundaries easily).
- **Regime:** Laminar flow ($Re \approx 700$).

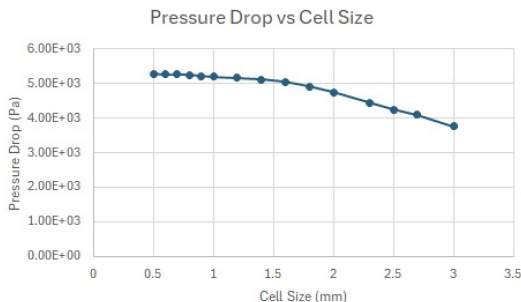


Figure: Grid convergence analysis (Converged at cell size 0.5).

LBM: Governing Equations

The LBM operates at a mesoscopic scale, tracking particle distribution functions f_i on a discrete lattice. The evolution consists of these steps:

1. Lattice Boltzmann Equation (LBE)

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = \Omega_i(f)$$

Collision Step (BGK Model):

$$\Omega_i(f) = -\frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)]$$

where τ is the relaxation time, directly linked to kinematic viscosity ν .

Macroscopic Reconstruction:

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{c}_i$$

Surrogate Architecture: Physics-Guided GPR

We replaced ANNs with **Gaussian Process Regressors (GPR)** to quantify uncertainty.

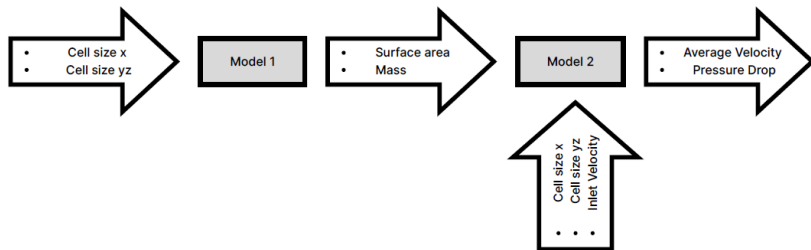


Figure: Two-stage architecture separating Geometric and Fluid predictions.

- **Model 1:** Inputs $(C_x, C_{yz}) \rightarrow$ Outputs (M, A_s) .
- **Model 2:** Inputs $(C_x, C_{yz}, v_{in}, M, A_s) \rightarrow$ Outputs $(\Delta p, v_{avg})$.

Data Generation: Active Learning

Instead of a full-factorial grid, we used an iterative approach:

- ① **Initial Set:** 75 Latin Hypercube Samples (LHS).
- ② **Active Learning Loop:** Added points based on an acquisition function balancing:
 - **Exploitation:** Relevance to the 125g mass constraint.
 - **Exploration:** High GPR uncertainty.

Efficiency

Total simulations used: **319** (compared to 787 in previous approaches).

Custom Acquisition Function

We designed a hybrid acquisition function to guide the Active Learning loop. It prioritizes points that are both **uncertain** and **relevant** to the constraints.

Mathematical Formulation

$$\alpha(\mathbf{x}) = \underbrace{[\tilde{\sigma}_M(\mathbf{x}) + \tilde{\sigma}_{A_s}(\mathbf{x})]}_{\text{Exploration}} \times \underbrace{\exp\left(-\frac{|\mu_M(\mathbf{x}) - M_{lim}|}{w}\right)}_{\text{Exploitation}}$$

Term Breakdown:

- **Exploration:** Adds normalized uncertainties ($\tilde{\sigma}$) from the Mass and Surface Area GPR models.
- **Exploitation:** A decay function centered at the constraint boundary ($M_{lim} = 125$ g). It favors points predicted to lie near the critical mass limit.

Optimization Pipeline

A three-stage hybrid strategy:

- 1 **Global Exploration:** Bayesian Optimization to find promising regions.
- 2 **Trust Region Refinement:** Added 50 LHS samples in the best region [14 – 16, 24 – 25] mm.
- 3 **Local Optimization:** Final constrained optimization within the trust region.

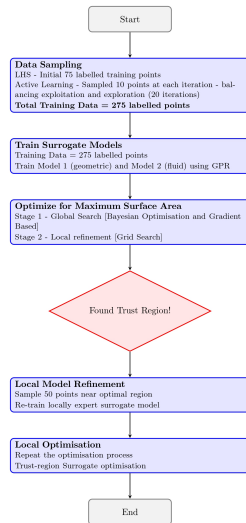


Figure: Optimization Flowchart

Final Optimized Design

Optimal Parameters:

- C_x : 15.36 mm
- C_{yz} : 24.86 mm

Performance:

- **Area:** 24,116 mm²
- **Mass:** 124.99 g (On limit)

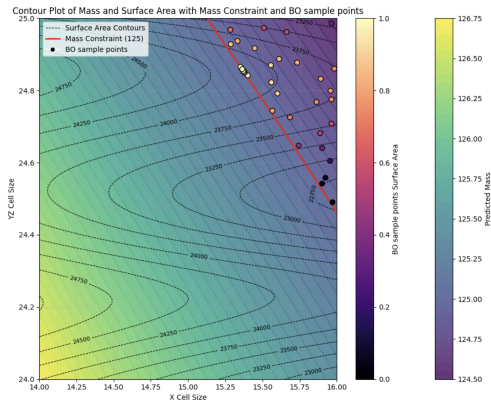


Figure: Contour plot.
Red line = 125g constraint.

Comparison: Mid-Sem vs. Final

The shift from ANNs/Full-Factorial to GPR/Active Learning yielded better results with less data.

Parameter	Mid-Sem (ANN)	Final (GPR)
Data Strategy	Full-Factorial	Active Learning
Data Points	787	319
C_x (mm)	21.78	15.36
C_{yz} (mm)	20.52	24.86
Mass (M)	124.99 g	124.99 g
Surface Area (A_s)	23,559 mm²	24,116 mm²

Summary & Conclusions

- **Framework:** Developed a robust surrogate-driven inverse design framework for TPMS heat exchangers.
- **Physics-Guided:** Separated geometry and flow physics in the surrogate architecture for better generalization.
- **Efficiency:** Reduced simulation time from weeks to hours using Active Learning.
- **Outcome:** Achieved a **2.36% increase** in surface area compared to the mid-semester baseline, while strictly adhering to mass and pressure constraints.

Thank You

Questions?